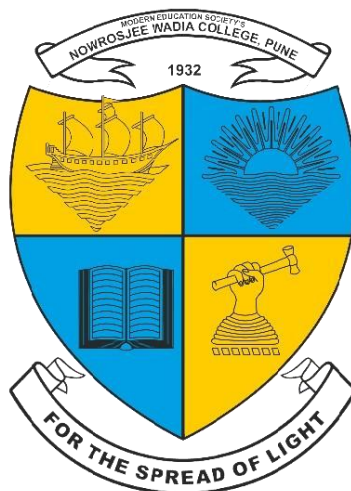




**Modern Education Society's
Nowrosjee Wadia College Pune - 01
Department of Biotechnology**

Topic of Research Project

**“Green Synthesis Characterization & Antimicrobial Analysis
of CuNPs from leaves of plant *Canavalia gladiata* (Jacq.)
DC.”**

Submitted By

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Roll No 09 M.Sc. II

2025-2026

For The Award of Degree

Masters of Science in Biotechnology

Research Guide

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CERTIFICATE

This is to certify that **Miss. Payal Dinkar Londhe** of M.Sc. II Biotechnology, Nowrosjee Wadia college, has successfully completed the project entitled **“Green Synthesis Characterization & Antimicrobial Analysis of plant *Canavalia gladiata* (Jacq.) DC”** for the course, PBTRP236 – Research Project towards the fulfilment of the syllabus of the master's degree in Biotechnology, as specified by the University of Pune, under the guidance of, Mr. Sagar D. Palwe Asst. Professor during the academic year 2025-2026.

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INTRODUCTION

Nanotechnology is an emerging field of science that deals with the synthesis, characterization, and application of materials at the nanoscale level (1–100 nm). Metal nanoparticles have gained significant attention due to their unique physical, chemical, optical, and biological properties compared to their bulk materials. Copper nanoparticles (CuNPs) are widely studied because of their excellent electrical conductivity, catalytic activity, antimicrobial properties, and cost-effectiveness compared to noble metals like gold and silver. However, conventional physical and chemical synthesis methods involve toxic chemicals, high energy consumption, and generate hazardous by-products, which are harmful to the environment and human health. Green synthesis of nanoparticles is an eco-friendly, cost-effective, and sustainable alternative. In this method, plant extracts act as reducing, stabilizing, and capping agents. Plant-mediated synthesis avoids the use of toxic chemicals and harsh conditions. *Canavalia gladiata* (Sword bean) is a medicinal plant rich in bioactive compounds such as flavonoids, phenols, alkaloids, proteins, and carbohydrates. These phytochemicals play a crucial role in reducing copper ions into copper nanoparticles and stabilizing them. The present study focuses on the green synthesis of copper nanoparticles using *Canavalia gladiata* leaf extract and their confirmation using UV-Visible spectroscopy (Iravani, 2011).

Nanotechnology has become an integral component of modern scientific research due to its ability to manipulate matter at the atomic and molecular scale, enabling the development of advanced materials with novel functionalities. Nanomaterials have demonstrated significant potential in overcoming limitations associated with conventional materials, particularly in areas requiring enhanced efficiency, sensitivity, and specificity. As a result, nanotechnology has found wide-ranging applications in medicine, pharmaceuticals, agriculture, energy, electronics, and environmental sciences. Metal nanoparticles are of particular interest because of their excellent physicochemical stability and multifunctional properties. Among them, copper-based nanoparticles represent a promising alternative to noble metal nanoparticles such as gold and silver due to their relatively lower cost, abundant availability, and comparable biological and catalytic activities. Copper sulphate-based nanoparticles, in particular, have gained

attention because copper ions play essential roles in biological systems and exhibit broad-spectrum antimicrobial activity against bacteria, fungi, and other pathogenic microorganisms. In recent years, the demand for sustainable and environmentally benign synthesis routes has increased due to growing concerns about ecological safety and human health. The excessive use of hazardous chemicals and energy-intensive processes in conventional nanoparticle synthesis has raised serious environmental and toxicological issues. Consequently, researchers are increasingly focusing on biologically inspired synthesis approaches that align with the principles of green chemistry. Plant-mediated green synthesis offers several advantages over microbial and chemical methods, including simplicity, rapid reaction rates, scalability, and reduced risk of contamination. Plant extracts serve as a rich source of secondary metabolites that not only facilitate the reduction of metal ions but also enhance the stability and biocompatibility of the synthesized nanoparticles. The use of plant extracts eliminates the need for external reducing and capping agents, thereby minimizing chemical waste and improving sustainability (Ahmed *et al.*, 2016).

Medicinal plants are especially valuable in green nanotechnology due to their therapeutic relevance and high phytochemical content. The incorporation of plant-derived bioactive compounds onto the surface of nanoparticles may further enhance their biological activity and functionality. This synergistic interaction between plant metabolites and metal nanoparticles has opened new avenues for the development of eco-friendly nanomaterials with improved efficacy (Singh *et al.*, 2018).

Canavalia gladiata, commonly known as sword bean, has been traditionally used in herbal medicine and is known to possess a diverse range of phytochemicals, including phenolic compounds, flavonoids, and proteins. These compounds are capable of acting as effective reducing and stabilizing agents during nanoparticle synthesis. Despite its medicinal importance, limited studies have explored the potential of *Canavalia gladiata* in the green synthesis of metal nanoparticles, particularly copper-based nanoparticles. Therefore, exploring *Canavalia gladiata* as a biological resource for nanoparticle synthesis contributes not only to sustainable nanotechnology but also to the value addition of medicinal

plants in advanced material science. The synthesis and characterization of copper sulphate nanoparticles using plant extracts may provide insights into developing low-cost, eco-friendly nanomaterials suitable for future biomedical and environmental applications (Vadivel & Janardhanan, 2002).

Canavalia gladiata (Jacq.) DC., commonly known as sword bean, gladiolus bean, or scimitar bean, is a perennial climbing leguminous plant belonging to the family Fabaceae. The genus *Canavalia* comprises approximately 48–60 species distributed mainly in tropical and subtropical regions of the world. Among them, *C. gladiata* is one of the most widely cultivated and scientifically investigated species due to its nutritional, medicinal, agricultural, and biotechnological significance. This plant is native to South and Southeast Asia, particularly India, Sri Lanka, Indonesia, and parts of China, and has been traditionally cultivated in tropical regions for centuries. In India, *Canavalia gladiata* is grown in several states including Maharashtra, Karnataka, Tamil Nadu, Andhra Pradesh, and Kerala, primarily as a food crop, fodder plant, green manure, and medicinal resource (Gan *et al.*, 2016).

The plant is characterized by its vigorous climbing habit, thick woody stems, large trifoliate leaves, and striking long sword-shaped pods, from which its common name “sword bean” is derived. The pods can grow up to 30–40 cm in length, containing large, hard seeds that vary in colour from white and pink to red and brown (Debbarma *et al.*, 2023).

From an agronomic perspective, *Canavalia gladiata* plays a significant role in sustainable agriculture. As a leguminous plant, it possesses the ability to form symbiotic associations with nitrogen-fixing *Rhizobium* bacteria, thereby improving soil fertility by enriching nitrogen content. For this reason, it is frequently used as a cover crop and green manure, particularly in tropical farming systems. The plant is highly adaptable and can tolerate poor soils, drought conditions, and high temperatures, making it valuable in regions with marginal agricultural land. Its dense foliage also helps in weed suppression and soil erosion control, contributing to ecological stability (More *et al.*, 2022).

The seeds of *Canavalia gladiata* are rich in proteins, carbohydrates, dietary fibre, and essential minerals such as calcium, iron, phosphorus, and potassium. Due to its high protein content, the sword bean is considered a potential alternative protein source, especially in developing countries facing food security challenges. However, raw seeds contain anti-nutritional factors, including protease inhibitors, lectins, and non-protein amino acids (such as canavanine). These compounds necessitate proper processing methods like soaking, boiling, fermentation, or roasting before consumption. Traditional food practices have long addressed these issues, demonstrating indigenous knowledge of safe utilization (Park, 2016).

Canavalia gladiata has a long history of use in traditional medicine systems, including Ayurveda, Siddha, and folk medicine. Different parts of the plant—seeds, leaves, roots, and pods—have been used to treat various ailments. Traditionally, the plant has been reported to exhibit antimicrobial, anti-inflammatory, antioxidant, antidiabetic, and anthelmintic properties. These medicinal properties are attributed to the presence of bioactive phytochemicals such as flavonoids, phenolic compounds, alkaloids, saponins, tannins, and lectins. From a biochemical standpoint, *Canavalia gladiata* is of particular interest due to the presence of unique seed proteins and lectins, which have been extensively studied for their biological activity and molecular interactions. Lectins isolated from *Canavalia* species are known for their carbohydrate-binding specificity and have applications in cell biology, immunology, and cancer research. The plant also serves as a model organism for studying plant defence mechanisms, as many of its secondary metabolites play roles in protecting against pests, pathogens, and environmental stress (Liu *et al.*, 2013).

Nanotechnology is an emerging interdisciplinary field that involves the manipulation of materials at the nanoscale level (1–100 nm), where unique physical, chemical, and biological properties are observed. These nanoscale materials exhibit enhanced reactivity, surface area, and biological interactions compared to their bulk counterparts, making them highly useful in biomedical applications such as drug delivery and antimicrobial therapy. nanoparticles, particularly metal nanoparticles such as copper, silver,

and gold, have gained significant attention due to their strong antimicrobial properties. These particles can interact with microbial cell membranes, disrupt cellular processes, and ultimately lead to cell death. Their small size allows them to penetrate microbial cells efficiently, making them potent antimicrobial agents. Green synthesis of nanoparticles is an eco-friendly and sustainable method that uses plant extracts, microorganisms, or biological systems for nanoparticle production. Plant-mediated synthesis is particularly advantageous due to its simplicity, cost-effectiveness, and reduced toxicity. Leaves of plants such as *Canavalia gladiata* contain phytochemicals that act as reducing and stabilizing agents in nanoparticle synthesis. The antimicrobial activity of nanoparticles is tested against various pathogenic microorganisms to evaluate their effectiveness. Common bacterial strains used include *Escherichia coli*, *Salmonella typhi*, and *Staphylococcus aureus*, which are responsible for a wide range of infections in humans. (Pelczar M., 2005)

Escherichia coli is a Gram-negative bacterium commonly found in the intestinal tract of humans and animals. While most strains are harmless, some can cause serious infections such as diarrhea, urinary tract infections, and septicaemia. Its outer membrane provides resistance to many antibiotics, making it a suitable candidate for antimicrobial studies. *Salmonella typhi* is the causative agent of typhoid fever, a serious systemic infection. It is also a Gram-negative bacterium and poses significant health challenges, especially in developing countries. The emergence of antibiotic-resistant strains has increased the need for alternative antimicrobial strategies such as nanoparticles. *Staphylococcus aureus* is a Gram-positive bacterium responsible for skin infections, pneumonia, and food poisoning. It is known for its ability to develop resistance against multiple antibiotics, including methicillin-resistant *S. aureus* (MRSA), making it a critical target for new antimicrobial agents. (Lowy F., 1998)

The mechanism of antimicrobial action of nanoparticles involves multiple pathways. These include generation of reactive oxygen species (ROS), disruption of cell membrane integrity, interaction with DNA and proteins, and inhibition of enzyme activity. Such multi-targeted mechanisms reduce the chances of microbial resistance development. Antibiotics such as Amoxicillin, Ampicillin, and

Levofloxacin are commonly used as standard drugs in antimicrobial studies for comparison. These antibiotics belong to different classes and act through different mechanisms, providing a reference to evaluate the effectiveness of nanoparticles. Amoxicillin and Ampicillin are β -lactam antibiotics that inhibit bacterial cell wall synthesis by targeting penicillin-binding proteins. They are effective mainly against Gram-positive bacteria but also show activity against some Gram-negative organisms. However, resistance due to β -lactamase enzymes is a major limitation. (Bush K., 2010)

Levofloxacin is a broad-spectrum fluoroquinolone antibiotic that inhibits bacterial DNA gyrase and topoisomerase IV, enzymes essential for DNA replication. It is effective against both Gram-positive and Gram-negative bacteria and is widely used in clinical practice. In antimicrobial assays, methods such as the agar well diffusion or disk diffusion technique are commonly employed. These methods measure the zone of inhibition produced by nanoparticles and antibiotics against test organisms, indicating their antimicrobial potency. The comparison between nanoparticles and conventional antibiotics helps in understanding their relative effectiveness. In many studies, nanoparticles have shown enhanced or synergistic effects when used in combination with antibiotics, indicating their potential role in overcoming antibiotic resistance. (Fayaz A., 2010)

Nanoparticles synthesized using plant extracts not only exhibit antimicrobial properties but also possess antioxidant and biocompatible characteristics. This makes them suitable for pharmaceutical and medical applications with minimal side effects. Overall, the study of antimicrobial activity of nanoparticles against pathogens like *E. coli*, *S. typhi*, and *S. aureus*, along with comparison using antibiotics such as Amoxicillin, Ampicillin, and Levofloxacin, provides valuable insights into developing new strategies to combat microbial resistance and improve therapeutic outcomes. Nanoparticles exhibit size-dependent properties that significantly influence their antimicrobial efficiency. As the particle size decreases, the surface area to volume ratio increases, enhancing their interaction with microbial cells. This increased surface reactivity allows nanoparticles to bind effectively with bacterial cell walls and membranes, leading to structural damage and leakage of cellular contents. The surface charge of nanoparticles also

plays a crucial role in antimicrobial activity. Positively charged nanoparticles show stronger interaction with negatively charged bacterial cell membranes, resulting in enhanced antibacterial effects. This electrostatic attraction facilitates better adhesion and penetration into microbial cells. Copper nanoparticles (CuNPs), in particular, are widely studied for their antimicrobial properties due to their cost-effectiveness and strong biocidal activity. Copper ions released from nanoparticles can interact with proteins and nucleic acids, causing cellular dysfunction and death. Additionally, copper generates oxidative stress inside microbial cells. Phytochemicals present in plant extracts such as flavonoids, alkaloids, and phenolic compounds play an important role in nanoparticle synthesis and stabilization. These compounds not only reduce metal ions into nanoparticles but also enhance their biological activity, including antimicrobial effects. The stability of nanoparticles is an important factor in determining their effectiveness. Aggregation of nanoparticles can reduce their active surface area and decrease antimicrobial activity. Therefore, proper stabilization using biological agents or capping molecules is essential to maintain their functional properties. The use of nanoparticles in combination with antibiotics has shown synergistic effects in many studies. This combination enhances the overall antibacterial activity and reduces the required dosage of antibiotics, thereby minimizing side effects and slowing down the development of resistance. (Liu Y., 2016)

Antibiotic resistance is a major global health concern caused by overuse and misuse of antibiotics. Bacteria develop resistance mechanisms such as enzyme production, efflux pumps, and mutation of target sites. Nanoparticles provide an alternative approach due to their multi-target mechanism of action. Gram-positive and Gram-negative bacteria show different responses to nanoparticles due to differences in their cell wall structure. Gram-negative bacteria have an outer membrane that acts as a barrier, while Gram-positive bacteria have a thick peptidoglycan layer. These structural differences influence nanoparticle penetration and effectiveness. (Silhavy T., 2010)

Characterization techniques such as UV-Visible spectroscopy, Scanning Electron Microscopy (SEM), and X-ray Diffraction (XRD) are used to confirm the formation and properties of nanoparticles. These

techniques help in determining particle size, shape, and crystalline nature, which are important for antimicrobial activity. Nanoparticles have wide applications beyond antimicrobial activity, including use in drug delivery, cancer therapy, biosensors, and environmental remediation. Their versatility and efficiency make them a promising tool in modern science and technology, especially in overcoming current biomedical challenges. (Bhattacharya R., 2008)



Fig. Fruit of *Canvalia gladiata*



Fig. Leaf of *Canvalia Gladiata*

RIVIEW OF LITERATURE

1. Early Botanical and Taxonomical Studies

Early scientific literature on *Canavalia gladiata* primarily focused on its taxonomy, morphology, and geographical distribution. Classical botanical surveys described *C. gladiata* as a robust perennial climber belonging to the Fabaceae family, characterized by trifoliate leaves, papilionaceous flowers, and elongated sword-shaped pods. Taxonomists distinguished *C. gladiata* from closely related species such as *Canavalia ensiformis* based on pod morphology, seed size, and growth habit. These foundational studies established the plant's classification and provided baseline botanical descriptions essential for later agronomic and biochemical research (Piper, 1913; Duke, 1981).

2. Agronomic and Soil Fertility Studies

Several agronomic studies have highlighted *Canavalia gladiata* as an important cover crop and green manure. Researchers reported its strong nitrogen-fixing ability through symbiotic association with *Rhizobium* species, contributing to improved soil nitrogen content and enhanced crop productivity. Studies conducted in tropical and subtropical regions demonstrated that cultivation of *C. gladiata* improves soil structure, reduces erosion, and suppresses weed growth. Its tolerance to drought, low soil fertility, and high temperature has been repeatedly emphasized, making it suitable for sustainable agriculture and marginal lands (Cherr *et al.* 2006; Samson, 1986;).

3. Nutritional Composition and Food Potential

A significant portion of literature focuses on the nutritional evaluation of *Canavalia gladiata* seeds. Researchers have reported high levels of crude protein, complex carbohydrates, and dietary fibre, along with essential minerals such as calcium, iron, and phosphorus. Comparative studies between *C. gladiata* and other legumes indicated that sword bean protein content is comparable to commonly consumed pulses.

However, studies also identified the presence of anti-nutritional factors like trypsin inhibitors, lectins, tannins, and canavanine. Research emphasized the importance of processing methods (soaking, boiling, fermentation, roasting) to reduce toxicity and improve digestibility. These findings support the plant's food potential while highlighting safety precautions (; Liener, 1994; Siddhuraju *et al.*, 1996).

4. Phytochemical Investigations

Phytochemical screening studies consistently reported bioactive compounds in *Canavalia gladiata*. Analyses revealed flavonoids, phenolics, saponins, alkaloids, glycosides, and tannins in various plant parts. Researchers suggested that these phytochemicals contribute to the plant's medicinal properties.

Solvent extraction, chromatography, and spectroscopic techniques were used to identify active constituents. The antioxidant potential of *C. gladiata* has been linked with its phenolic and flavonoid content (Kumari & Rao, 2015; Harborne, 1998).

5. Lectins and Protein-Based Studies

Lectins from *Canavalia* species, including *C. gladiata*, have been widely studied due to their carbohydrate-binding specificity. These proteins show hemagglutinating activity and stability over various pH and temperature ranges.

They have applications in immunology, cancer biology, and cell surface studies. Some research also explored insecticidal and antimicrobial roles, indicating defensive functions in plants (Sharon & Lis, 2004 Moreira & Cavada, 1984).

6. Pharmacological and Medicinal Studies

Traditional medicine documented *C. gladiata* for treating inflammation, digestive issues, diabetes, and infections. Modern experimental studies demonstrated antimicrobial, antioxidant, anti-inflammatory, antidiabetic, and anthelmintic activities in extracts.

These effects are attributed to phytochemicals like flavonoids and saponins. However, authors have noted that further clinical evaluation is needed (Balasubramanian *et al.*, 2014; Pugalenti *et al.*, 2005).

7. Toxicological and Anti-Nutritional Studies

Studies highlighted toxicological risks of raw *C. gladiata* seeds due to antinutritional compounds like canavanine and lectins. Animal studies showed adverse effects when consumed without proper processing. Therefore, researchers strongly recommend techniques like boiling, roasting, soaking, and fermentation to eliminate toxicity (Goulet & Awori, 2003 Rosenthal, 2001).

8. Nanoparticles as Antimicrobial Agents

Nanotechnology has been extensively explored for antimicrobial applications due to the unique physicochemical properties of nanoparticles such as high surface area to volume ratio, enhanced reactivity, and ability to interact at the molecular level. Early studies demonstrated that metal nanoparticles like silver, copper, and gold exhibit strong antibacterial activity against a wide range of microorganisms including both Gram-positive and Gram-negative bacteria. These nanoparticles attach to the microbial cell surface, penetrate inside, and disrupt vital cellular functions such as respiration and nutrient transport. Due to these multi-target mechanisms, nanoparticles are considered promising alternatives to conventional antibiotics, especially in overcoming drug resistance. (Rai M., 2009)

2. Green Synthesis of Nanoparticles

Green synthesis of nanoparticles using plant extracts has gained attention as a sustainable, eco-friendly, and cost-effective approach compared to chemical and physical methods. Plant extracts contain bioactive compounds such as flavonoids, phenols, terpenoids, and alkaloids which act as both reducing agents (converting metal ions into nanoparticles) and stabilizing agents (preventing aggregation). This method eliminates the use of hazardous chemicals, making the nanoparticles more biocompatible and suitable for biomedical applications. Additionally, plant-mediated nanoparticles often show enhanced antimicrobial activity due to the synergistic effect of phytochemicals. (Iravani S., 2011)

3. Antimicrobial Activity of Copper Nanoparticles

Copper nanoparticles (CuNPs) have shown significant antibacterial activity due to their ability to release copper ions and generate reactive oxygen species (ROS). These ROS cause oxidative stress inside bacterial cells, damaging proteins, lipids, and DNA. CuNPs also disrupt the integrity of bacterial cell membranes, leading to leakage of intracellular contents and eventual cell death. Compared to other metal nanoparticles, copper is more economical and widely available, making it a practical choice for antimicrobial applications in research and medicine. (Ren G., 2009)

4. Effect of Nanoparticles on Escherichia coli

Escherichia coli, a Gram-negative bacterium, possesses an outer membrane rich in lipopolysaccharides which acts as a protective barrier against many antibiotics. However, nanoparticles can overcome this barrier due to their small size and high reactivity. They attach to the bacterial surface, penetrate the membrane, and disrupt cellular structures. This leads to leakage of essential biomolecules such as proteins and nucleic acids, ultimately inhibiting bacterial growth and causing cell death. Thus, *E. coli* serves as an important model organism in antimicrobial studies. (Stoimenov P., 2002)

5. Effect of Nanoparticles on Staphylococcus aureus

Staphylococcus aureus is a Gram-positive bacterium characterized by a thick peptidoglycan layer in its cell wall. Despite this structural barrier, nanoparticles can interact with the cell wall and inhibit its synthesis. They also interfere with protein function and enzymatic activity within the bacterial cell. Nanoparticles may bind to thiol groups in proteins, leading to enzyme inactivation. This results in impaired metabolic activity and eventual bacterial death. Their effectiveness against antibiotic-resistant strains like MRSA makes them highly significant. (Hajipour M., 2012)

6. Effect of Nanoparticles on Salmonella typhi

Salmonella typhi, the causative agent of typhoid fever, is a Gram-negative pathogen that poses serious health risks. Nanoparticles exert antibacterial effects by generating reactive oxygen species and interfering with essential metabolic pathways. They can also damage DNA and inhibit replication processes, thereby preventing bacterial proliferation. Due to increasing antibiotic resistance in *S. typhi*, nanoparticles are being explored as alternative therapeutic agents for controlling such infections. (Crump J., 2015)

7. Comparison with Conventional Antibiotics

Comparative studies between nanoparticles and conventional antibiotics such as Amoxicillin and Ampicillin have shown that nanoparticles often exhibit equal or even superior antibacterial activity. While antibiotics typically target specific cellular pathways, nanoparticles act through multiple mechanisms, reducing the chances of resistance development. Moreover, nanoparticles can remain effective against antibiotic-resistant strains, making them valuable in modern antimicrobial therapy. (Bush K., 2010)

8. Role of Levofloxacin in Antimicrobial Studies

Levofloxacin is a broad-spectrum fluoroquinolone antibiotic that acts by inhibiting bacterial enzymes DNA gyrase and topoisomerase IV, which are essential for DNA replication and cell division. This leads to inhibition of bacterial growth and cell death. When used in combination with nanoparticles, a synergistic effect is observed, where nanoparticles enhance drug penetration into bacterial cells, thereby increasing the overall antimicrobial efficacy. (Hooper D., 2001)

9. Methods for Antimicrobial Activity Testing

Agar well diffusion and disk diffusion methods are widely used techniques to evaluate antimicrobial activity. In these methods, nanoparticles or antibiotics are introduced into wells or discs placed on agar plates inoculated with bacteria. After incubation, the zone of inhibition around the well or disc indicates the effectiveness of the antimicrobial agent. Larger zones indicate stronger antibacterial activity. These methods are simple, reliable, and widely accepted in microbiological studies. (Bauer A., 1966)

10. Influence of Nanoparticle Size and Shape

The size and shape of nanoparticles play a crucial role in determining their antimicrobial efficiency. Smaller nanoparticles have a larger surface area, allowing greater interaction with bacterial cells. Spherical nanoparticles are generally more effective due to uniform distribution and better penetration. Shape-dependent activity influences how nanoparticles attach to and disrupt microbial membranes, making it an important factor in nanoparticle design. (Pal S., 2007)

11. Nanoparticles and Biofilm Disruption

Bacterial biofilms are structured communities of microorganisms embedded in a protective extracellular matrix, making them highly resistant to antibiotics. Nanoparticles can penetrate this biofilm matrix and

disrupt its structure. They break down the extracellular polymeric substances and kill the bacteria within the biofilm. This property makes nanoparticles highly useful in treating chronic and persistent infections. (Singh R., 2017)

12. Synergistic Effect with Antibiotics

Studies have shown that nanoparticles can work synergistically with antibiotics to enhance antibacterial activity. Nanoparticles increase the permeability of bacterial cell membranes, allowing4 greater uptake of antibiotics. This combination reduces the required dose of antibiotics and minimizes side effects. It also helps in overcoming antibiotic resistance, making combination therapy highly effective.

13. Toxicity and Biocompatibility of Nanoparticles

Green synthesized nanoparticles are considered safer compared to chemically synthesized ones due to their lower toxicity and higher biocompatibility. The presence of plant-derived capping agents reduces harmful effects on human cells. This makes them suitable for medical applications such as drug delivery, wound healing, and antimicrobial coatings. However, proper toxicity evaluation is still essential before clinical use.

14. Nanoparticles in Targeted Drug Delivery

Nanoparticles are increasingly used in targeted drug delivery systems, where they deliver antimicrobial agents directly to the site of infection. This targeted approach increases drug concentration at the infection site while reducing systemic side effects. It also improves therapeutic efficiency and reduces treatment duration. Such advanced applications highlight the future potential of nanotechnology in medicine.

SIGNIFICANCE OF THE STUDY

Using *Canavalia gladiata* leaf extract to synthesize copper nanoparticles (CuNPs) is significant because it combines green chemistry with nanotechnology.

- The leaves contain natural phytochemicals that act as reducing and stabilizing agents, enabling nanoparticle formation without toxic chemicals. This eco-friendly method reduces cost, energy use, and environmental hazards.
- The resulting CuNPs often show strong antimicrobial, antioxidant, and catalytic activities due to their nanoscale size and enhanced surface area.
- These properties make them valuable for biomedical applications, agricultural disease control, wastewater treatment, and sensor development.
- Green synthesis of nanoparticles is eco-friendly and cost-effective. It avoids toxic chemicals. It is safe for biomedical use. It promotes sustainable research.
- The study compares nanoparticles with antibiotics like Amoxicillin, Ampicillin, and Levofloxacin. It shows their effectiveness against bacteria. It helps in improving drug performance. It supports better treatment strategies.
- Nanoparticles act by multiple mechanisms like ROS generation and membrane damage. This reduces chances of resistance. It improves antibacterial action. It explains their working clearly.

- Nanoparticles have applications in drug delivery and infection control. They can be used in medical devices. They improve treatment efficiency. They enhance healthcare safety.
- This study helps in developing new nanoparticle-based therapies. It supports future research in nanotechnology. It addresses global health problems. It advances modern medicine

Aim and Objectives

- Collection of *Canavalia gladiata* plant.
- Authentication of *Canavalia gladiata*.
- Preparation of aqueous leaf extract.
- Green synthesis of CuNPs.
- Characterization of CuNPs.
- To study antimicrobial activity

PLAN OF WORK

Selection and collection of plant material



Preparation of leaf powder



Preparation of aqueous plant extract



Preparation of copper sulphate solution



Preparation of sodium hydroxide solution



Green synthesis of copper nanoparticles



Purification and calcination of nanoparticles



UV-Visible spectroscopic analysis



Preparation of culture media



Antimicrobial activity testing

MATERIALS REQUIRED

Chemicals

- Copper sulphate (CuSO_4)
- Sodium hydroxide (NaOH)
- Deionized water
- Plant Material
- Fresh leaves of *Canavalia gladiata*

Glassware

- Beakers
- Measuring cylinders
- Conical flasks

Instruments

- Centrifuge & centrifuge tubes
- Hot Air Oven
- Centrifuge
- UV-Visible spectrophotometer & cuvette
- Magnetic stirrer with hot plate

Materials Required for antimicrobial testing

- Autoclave
- Micropipette tips and tip box
- Test tubes
- Conical flasks (sterile distilled water)
- Beakers
- Spreader (L-shape)
- Petri plates
- Eppendorf tubes
- Nutrient agar medium
- Nutrient broth
- Nanoparticle (CuNPs) powder/solution
- Distilled water (d/w)
- Alcohol (for sterilization)
- Gloves
- Cotton
- Marker
- Bunsen burner
- Incubator
- Laminar airflow cabinet
- Microbial Cultures
- Escherichia coli
- Salmonella typhi
- Staphylococcus aureus
- Antibiotics Used

- Amoxicillin
- Ampicillin
- Levofloxacin

PROCEDURE

A. Selection and Collection of Plant Material:

- Healthy leaves of *Canavalia gladiata* were selected and collected manually. **B. Washing:**
- Leaves were thoroughly washed with tap water followed by deionized water to remove dust and impurities.

C. Shade Drying:

- Washed leaves were shade-dried at room temperature for several days to remove moisture without destroying phytochemicals. **D. Grinding:**

- Dried leaves were ground into a fine powder using a clean grinder. **E. Preparation of Plant**

Extract:

- 20 g of leaf powder was added to 100 ml deionized water
- Heated at 60°C for 3 hours
- Allowed to cool at room temperature
- Filtered using muslin cloth
- Centrifuged at 10,000 rpm for 10 minutes
- Clear supernatant was collected and stored in a cool place **F. Preparation of Metal Salt**

Solution:

- 0.2495 g of CuSO₄ was dissolved in 200 ml deionized water to prepare 5 mM solution **G.**

Preparation of NaOH Solution:

- 16 g NaOH dissolved in 200 ml deionized water to prepare 2N NaOH

H. Green Synthesis of Copper Nanoparticles:

- Copper sulphate solution was added dropwise into the plant extract

- pH was adjusted to basic using 2N NaOH
 - Continuous stirring was maintained
 - Colour change was observed indicating nanoparticle formation
 - Reaction mixture was kept overnight
- I. Purification of Nanoparticles:**
- Centrifuged at 10,000 rpm for 10 minutes
 - Pellet collected
 - Washed with deionized water
- J. Calcination:**
- Dried pellet was calcined at 300°C for 4 hours
 - Fine copper nanoparticle obtained
- K. Characterization of CuNPs:**
- Colour change indicates synthesis of CuNPs.
 - UV Visible spectroscopy.

Green Synthesis and Characterization of CuNPs from leaves of plant *Canavalia gladiata*



Fig. Fresh Leaves



Fig. Shade dry Leaves



Fig. Fine Powder



Fig. Heated on Magnetic Stirrer



Fig. Preparation of Salt Solution



Fig. Plant crude extract

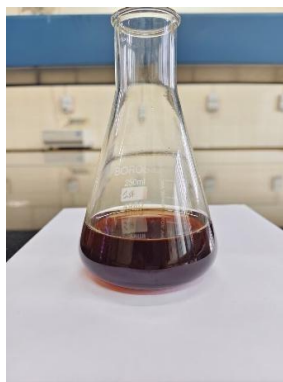


Fig. Extract changes copper



Fig. Nanoparticles

Procedure For Antimicrobial Activity

- All glassware (conical flasks, test tubes, petri plates) and media were sterilized in an autoclave at 121°C for 15–20 minutes. The laminar airflow cabinet was cleaned with 70% alcohol and exposed to UV light before starting the experiment.
- Nutrient broth was prepared and sterilized. It was inoculated separately with *Escherichia coli*, *Salmonella typhi*, and *Staphylococcus aureus* under aseptic conditions. The cultures were incubated at 37°C for 24 hours to obtain active bacterial growth.
- Nutrient agar medium was prepared, sterilized, and poured into sterile petri plates under laminar airflow. The plates were allowed to solidify at room temperature.
- The agar plates were labelled and each bacterial culture was spread uniformly using a sterile L-shaped spreader to obtain a lawn culture.
- Wells were made in the agar plates using a sterile cork borer. Proper distance was maintained between wells and labelling was done for different samples.
- **Preparation of Nanoparticle Samples**
- Nanoparticle (CuNPs) samples were prepared by adding different volumes of nanoparticle solution, i.e., 0.003 gm, 0.005 gm, 0.007 gm, and 0.01 gm, into 2 ml of sterile distilled water separately.
- These different concentrations were used to study the concentration-dependent antimicrobial activity.
- **Preparation of Antibiotic Samples**
- Antibiotic solutions were prepared by dissolving 0.5 mg antibiotic powder (Amoxicillin, Ampicillin, Levofloxacin) in 2 ml of sterile distilled water.

- The solutions were mixed properly to obtain uniform concentration for comparison.
- Required volume of each nanoparticle concentration was added into the respective wells using a micropipette. Antibiotic solutions were also added into separate wells. One well was maintained as control by adding only sterile distilled water.
- The plates were kept undisturbed for 10–15 minutes to allow proper diffusion of nanoparticles and antibiotics into the agar medium.
- The plates were incubated at 37°C for 24–48 hours in an inverted position.
- After incubation, the plates were observed for clear zones around the wells, indicating inhibition of bacterial growth.
- The zone of inhibition was measured in and the antimicrobial activity of different nanoparticle concentrations were compared with standard antibiotics.
- All results were recorded systematically and analysed to determine the effectiveness of nanoparticles against *E. coli*, *S. typhi*, and *S. aureus*.



fig. bacterial culture



fig. nanoparticle concentration



Fig. preparing plate

OBSERVATION & RESULT

UV-VISIBLE SPECTROSCOPY ANALYSIS:

- UV-Visible spectroscopy was used to confirm the formation of copper nanoparticles. The absorbance spectrum showed a characteristic peak due to surface plasmon resonance of copper nanoparticles, confirming successful synthesis.
- Successful synthesis of copper nanoparticles was observed.
- Fine copper nanoparticle obtained after calcination.
- Colour change from light blue to dark brown confirmed formation of CuNPs.
- UV-Visible spectrum showed characteristic absorption peak at **285nm** wave length.
- Further characterization will be performed Viz. XRD, FTIR and SEM EDS.

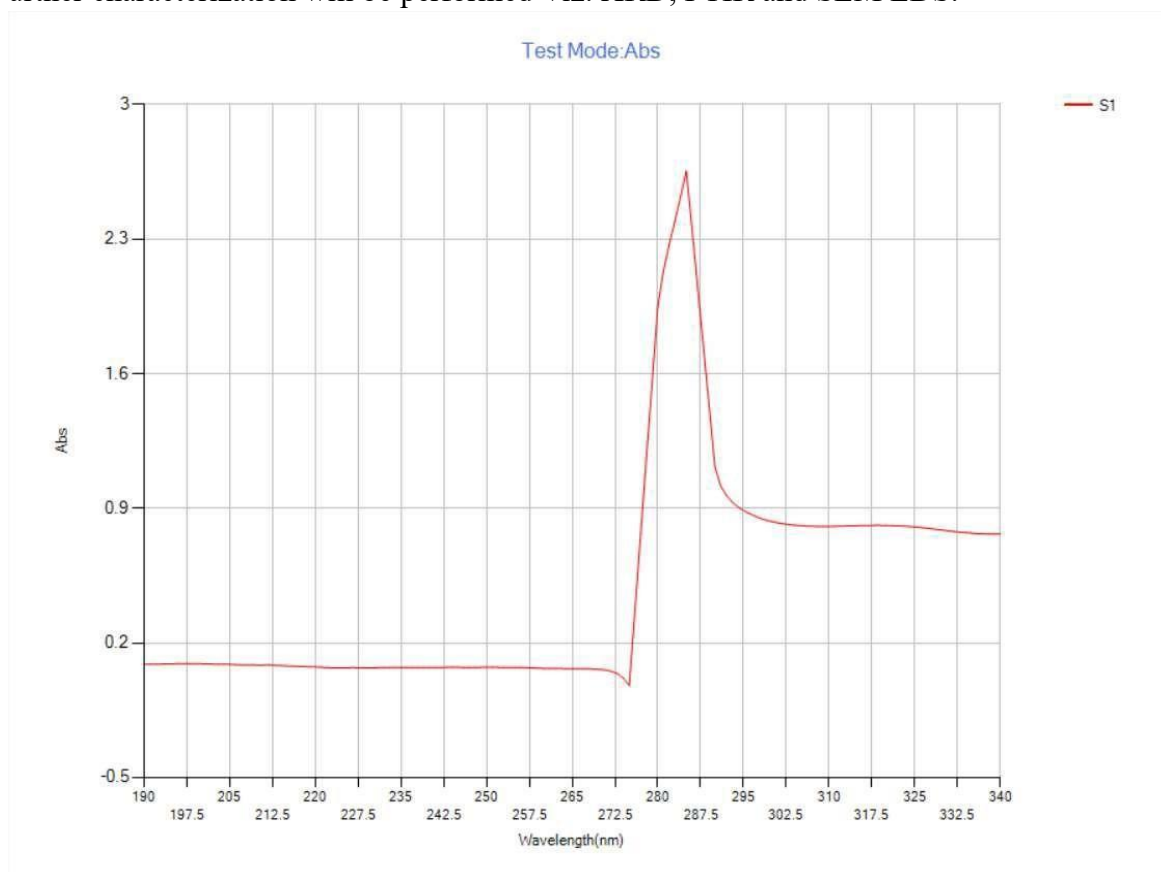


Fig UV–Visible Absorption Spectrum of Synthesized Cu Nanoparticles From the leaves of *Canvalia Gladiata* (Jacq.) DC.

Ultra Violet Visible Spectroscopy Data Sheet

The following data is generated in UV Visible Spectrum Analysis of Copper Nanoparticles synthesized from Leaves Extract of Plant *Canvalia Gladiata* (Jacq.) DC.

Mode: Spectrums can

Starting Wavelength(nm):340.0

End Wavelength(nm):190.0

Scan

Interval(nm):1.0

Display: Abs

High Limit:3

Low Limit: -0.5

ID	Wavelength(nm)	Abs	%T	Energy	Energy (100)	Energy (0)	Gain
1	340	0.7658	17.15	8448	683	3315	7
2	339	0.7662	17.13	8140	1339	1902	6
3	338	0.7709	16.95	8167	1354	1902	6
4	337	0.776	16.75	8188	1369	1902	6
5	336	0.7798	16.6	8202	1383	1902	6
6	335	0.7839	16.45	8206	1394	1902	6
7	334	0.7882	16.29	8213	1405	1902	6
8	333	0.7927	16.12	8219	1417	1902	6
9	332	0.7965	15.98	8228	1428	1902	6
10	331	0.8004	15.84	8242	1439	1902	6
11	330	0.8034	15.72	8229	1446	1902	6
12	329	0.8058	15.64	8224	1452	1902	6
13	328	0.8079	15.56	8219	1458	1902	6
14	327	0.8096	15.5	8216	1463	1902	6
15	326	0.8108	15.46	8220	1471	1902	6
16	325	0.8121	15.41	8188	1470	1902	6
17	324	0.813	15.38	8155	1468	1902	6
18	323	0.812	15.42	8109	1463	1902	6
19	322	0.8111	15.45	8063	1458	1902	6
20	321	0.8114	15.44	8004	1449	1902	6
21	320	0.8103	15.48	7916	1435	1902	6
22	319	0.8101	15.49	7814	1418	1902	6
23	318	0.8083	15.55	7718	1401	1902	6
24	317	0.8067	15.6	7599	1378	1902	6
25	316	0.8064	15.62	7460	1352	1902	6
26	315	0.8047	15.68	7306	1322	1902	6
27	314	0.8042	15.7	7138	1289	1902	6
28	313	0.8033	15.73	6956	1254	1902	6

29	312	0.8032	15.73	6752	1214	1902	6
30	311	0.8044	15.69	6523	1169	1902	6
31	310	0.8056	15.65	6276	1120	1902	6
32	309	0.8078	15.57	6016	1068	1902	6
33	308	0.8107	15.46	5740	1011	1902	6
34	307	0.8142	15.34	5459	949	1902	6
35	306	0.8185	15.19	5173	894	1902	6
36	305	0.8248	14.97	4851	831	1902	6
37	304	0.8267	14.9	7971	768	3315	7
38	303	0.8356	14.6	7471	706	3315	7
39	302	0.8447	14.3	6988	642	3315	7
40	301	0.8557	13.94	6505	582	3315	7
41	300	0.8687	13.53	6034	521	3315	7
42	299	0.8846	13.04	5585	461	3315	7
43	298	0.9025	12.52	5169	403	3315	7
44	297	0.9331	11.66	8013	350	5582	8
45	296	0.9494	11.23	7516	301	5582	8
46	295	0.9834	10.39	7035	257	5582	8
47	294	1.0251	9.44	6631	219	5582	8
48	293	1.0858	8.21	6289	185	5582	8
49	292	1.1845	6.54	6010	157	5582	8
50	291	1.3545	4.42	5796	134	5582	8
51	290	1.8763	1.33	5628	116	5582	8
53	288	4	0.01	5397	89	5582	8
54	287	4	0.01	5322	80	5582	8
55	286	4	0.01	5275	74	5582	8
56	285	4	0.01	5233	69	5582	8
57	284	-0.301	200	5211	66	5582	8
		-					
58	283	0.1784	150.81	5207	65	5582	8
		-					
59	282	0.0471	111.46	5205	63	5582	8
		-					
60	281	0.0069	101.6	5189	63	5582	8
61	280	0.0277	93.82	5190	62	5582	8
62	279	0.049	89.33	5192	62	5582	8
63	278	0.05	89.14	5187	62	5582	8
64	277	0.0674	85.62	5195	62	5582	8
65	276	0.0576	87.58	5191	62	5582	8
66	275	0.0674	85.63	5194	62	5582	8
67	274	0.0661	85.89	5190	62	5582	8
68	273	0.0596	87.18	5187	62	5582	8
69	272	0.0746	84.23	5192	62	5582	8
70	271	0.0575	87.59	5189	62	5582	8
71	270	0.0673	85.65	5193	62	5582	8
72	269	0.0679	85.52	5186	62	5582	8
73	268	0.0694	85.23	5193	62	5582	8
74	267	0.0573	87.64	5182	62	5582	8

75	266	0.0662	85.87	5192	62	5582	8
76	265	0.0682	85.47	5190	62	5582	8
77	264	0.067	85.7	5188	62	5582	8
78	263	0.0692	85.27	5190	62	5582	8
79	262	0.0702	85.08	5189	62	5582	8
80	261	0.068	85.5	5188	62	5582	8
81	260	0.0722	84.69	5188	62	5582	8
101	239	0.0807	83.04	5214	62	5582	8
102	238	0.0671	85.68	5208	62	5582	8
103	237	0.0764	83.86	5215	62	5582	8
104	236	0.0711	84.9	5216	62	5582	8
105	235	0.0742	84.29	5215	62	5582	8
106	234	0.0714	84.83	5222	62	5582	8
107	233	0.07	85.11	5218	62	5582	8
108	232	0.056	87.9	5209	62	5582	8
109	231	0.0701	85.09	5218	62	5582	8
110	230	0.067	85.71	5222	62	5582	8
111	229	0.0844	82.33	5228	62	5582	8
112	228	0.0656	85.97	5219	62	5582	8
113	227	0.0816	82.87	5234	62	5582	8
114	226	0.0713	84.86	5220	62	5582	8
115	225	0.0702	85.07	5221	62	5582	8
116	224	0.0753	84.08	5228	62	5582	8
117	223	0.0657	85.96	5221	62	5582	8
118	222	0.0644	86.21	5219	62	5582	8
119	221	0.0862	81.99	5225	62	5582	8
120	220	0.0839	82.43	5224	62	5582	8
121	219	0.0852	82.18	5226	62	5582	8
122	218	0.0871	81.83	5223	62	5582	8
123	217	0.088	81.67	5221	62	5582	8
124	216	0.0837	82.47	5222	62	5582	8
125	215	0.0781	83.55	5221	62	5582	8
126	214	0.0916	80.99	5224	62	5582	8
127	213	0.0799	83.2	5217	62	5582	8
128	212	0.091	81.11	5219	62	5582	8
129	211	0.0797	83.23	5215	62	5582	8
130	210	0.0794	83.29	5212	62	5582	8
131	209	0.094	80.53	5218	62	5582	8
132	133	0.0851	82.2	5215	62	5582	8
134	207	0.086	82.04	5213	62	5582	8
135	206	0.0988	79.65	5222	62	5582	8
136	205	0.0944	80.46	5213	62	5582	8
137	204	0.0985	79.7	5220	62	5582	8
138	203	0.0864	81.96	5207	62	5582	8
139	202	0.0953	80.29	5212	62	5582	8
140	201	0.0904	81.2	5206	62	5582	8
141	200	0.0837	82.48	5201	62	5582	8
142	199	0.0975	79.89	5205	62	5582	8

143	198	0.0889	81.48	5202	62	5582	8
144	197	0.0939	80.55	5201	62	5582	8
145	196	0.0911	81.08	5194	62	5582	8
146	195	0.0834	82.53	5188	62	5582	8
147	194	0.0939	80.55	5193	62	5582	8
148	193	0.0797	83.23	5181	62	5582	8
149	192	0.0852	82.19	5185	61	5582	8
150	191	0.0835	82.51	5178	61	5582	8
151	190	0.0949	80.38	5184	61	5582	8

Antimicrobial Test Result

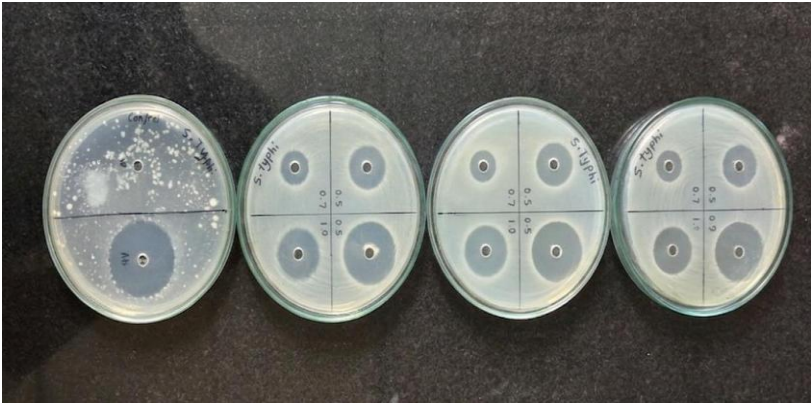
Fig. *E.coli*



Fig. *S.auresus*



Fig. *S.typhi*



Zone of inhibition

N o .	Concentration	E.coli	S.auresus	S.typhi
1	Control	22mm	28mm	24mm
2	0.3	n.d	14mm	12mm
3	0.5	8mm	17mm	15mm
4	0.7	10mm	21mm	17mm
5	1	12mm	24mm	18mm

CONCLUSION

- The present study confirms the successful green synthesis of copper nanoparticles using *Canavalia gladiata* leaf extract. The method is simple, cost-effective, and environmentally friendly, as it avoids the use of toxic chemicals. Phytochemicals present in the plant extract act as reducing and stabilizing agents for the formation of copper nanoparticles.
- The synthesis was confirmed by a visible green colour to copper colour change and further verified using UV-Visible spectroscopy, which showed a characteristic absorption peak between 280 nm to 320 nm. The purification and calcination steps produced stable copper nanoparticle, and alkaline pH was found to support nanoparticle stability and greater synthesis.
- Overall, plant-based green synthesis is a safe and sustainable approach with potential applications in biomedical, agricultural, catalytic, and environmental fields.
- Further study and characterization is needed to study the other possible perspectives in material science and its applications in various fields.
- Nanoparticles showed effective antimicrobial activity comparable to standard antibiotics against tested bacteria.

Future Perspective

- Using *Canavalia gladiata* leaves to produce copper nanoparticles is a promising area for future research.
- The leaves naturally contain compounds that can reduce and stabilize copper ions, allowing nanoparticles to form without using harmful chemicals.
- In the future, these plant-based nanoparticles can be improved by controlling their size and shape so that they work better as antimicrobial and antioxidant agents. They may be used for treating infections, coating medical tools, or making safer disinfectants.
- Their catalytic ability also makes them useful in water purification and pollution control. With more studies, *C. gladiata* leaf-derived copper nanoparticles can become an affordable, ecofriendly option for medical, agricultural, and industrial applications.

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